

Code:

```
#include <stdio.h> #include <stdlib.h>

struct Flat { long long c; // y - x long long h; // population };

int cmp(const void *a, const void *b) { struct Flat *f1 = (struct Flat *)a; struct Flat *f2 =
(struct Flat *)b; if (f1->c < f2->c) return -1; if (f1->c > f2->c) return 1; return 0; }

int main() { int t; if (scanf("%d", &t) != 1) return 0;

while (t--) {
    int n;
    if (scanf("%d", &n) != 1) break;

    struct Flat flats[n];
    long long total_pop = 0;

    for (int i = 0; i < n; i++) {
        long long x, y, h;
        scanf("%lld %lld %lld", &x, &y, &h);
        flats[i].c = y - x;
        flats[i].h = h;
        total_pop += h;
    }

    if (total_pop % 2 != 0) {
        printf("NO\n");
        continue;
    }

    qsort(flats, n, sizeof(struct Flat), cmp);

    long long pref[n];
    long long cur = 0;
    for (int i = 0; i < n; i++) {
        cur += flats[i].h;
        pref[i] = cur;
    }

    long long target = total_pop / 2;
    int possible = 0;
```

```

int l = 0, r = n - 2;
while (l <= r) {
    int mid = (l + r) / 2;
    if (pref[mid] == target) {
        if (flats[mid].c != flats[mid + 1].c) {
            possible = 1;
        }
        break;
    } else if (pref[mid] < target) {
        l = mid + 1;
    } else {
        r = mid - 1;
    }
}

if (possible) {
    printf("YES\n");
} else {
    printf("NO\n");
}
}

return 0;

}

```

Input:

```

3
3
-2 1 1
-1 1 3
1 -1 4
3
-2 1 1
-1 1 1
1 -1 4
4

```

-2 1 1
-1 1 2
1 -1 4
3 1 5

Output:

YES
NO
NO